

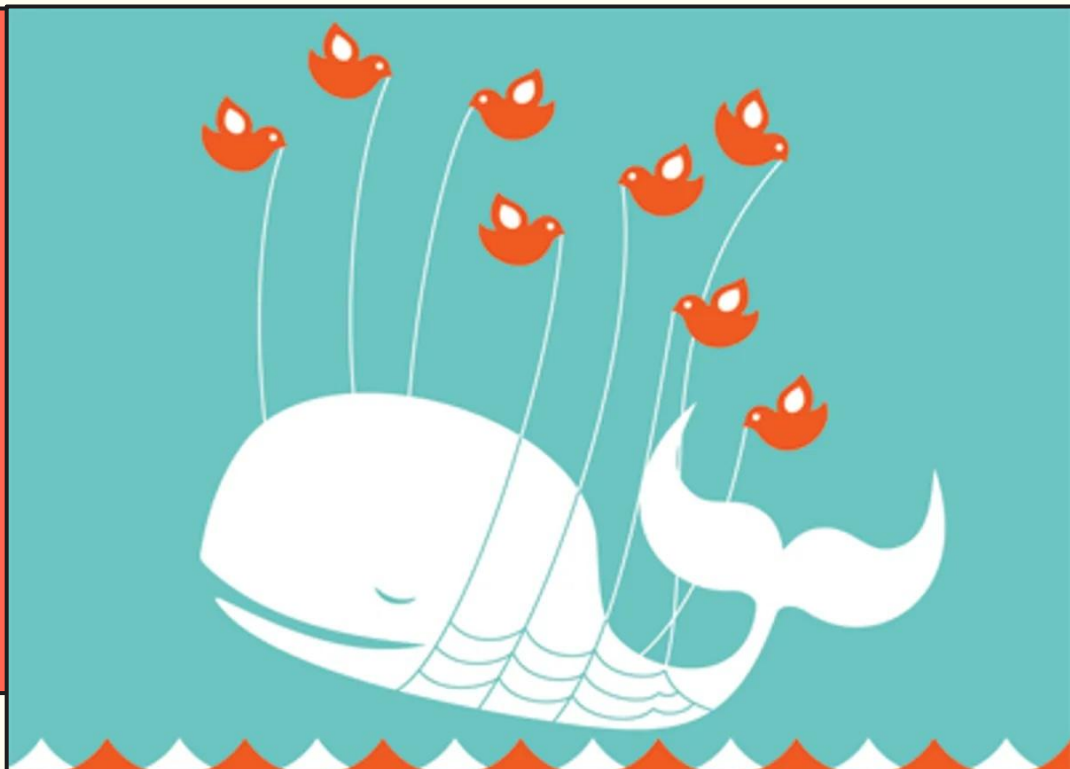
Richard Rose
HUFS / Yonsei

Validating Azure Pronunciation Assessment

Live Speech Test Demo

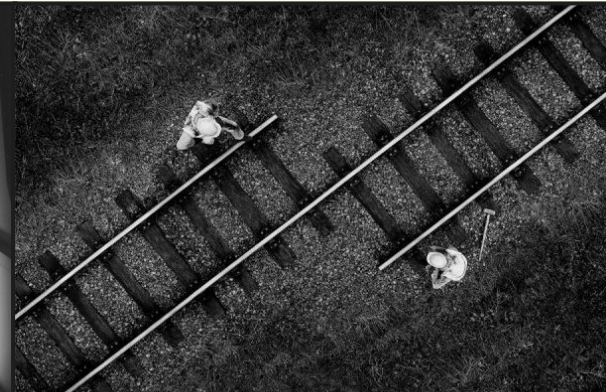
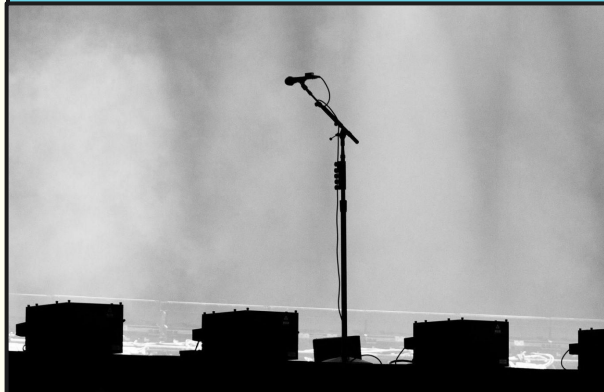
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- Choose any text
- I'll read aloud and test my speech
- [Demo Link Here!](#)
- Later we can check the analysis PDF
- You can also test your own speech



Presentation Overview

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The Problem & Potential Solutions

Neural network speech assessment is becoming more prevalent, but validation of these tools remains incomplete.

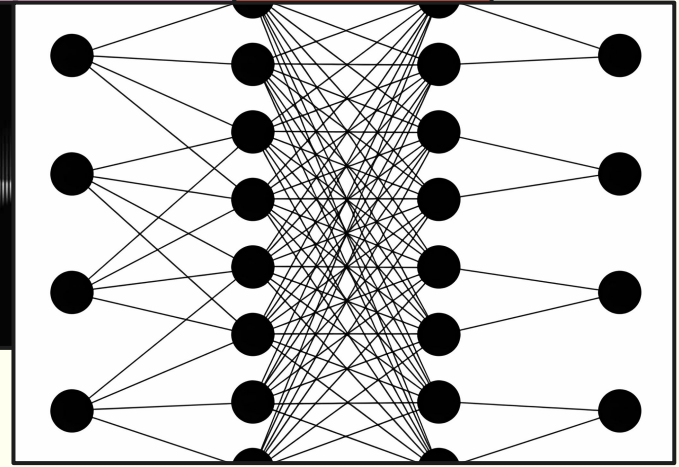
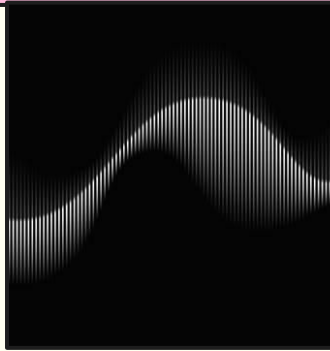
Methods & Measurement

ICNALE corpus ($n = 3,510$)
& classroom data ($n = 66$)
Comparisons of Azure scores with linguistic indices and human ratings

Results & Future Directions

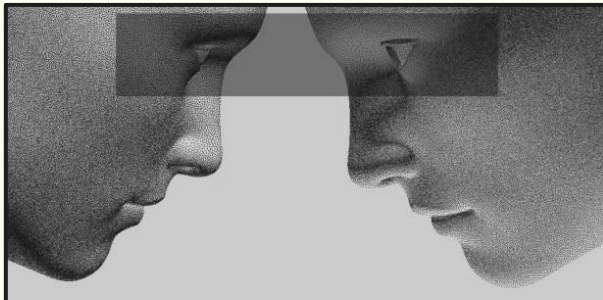
Strong rank-order alignment
but large calibration differences

The Problem & Potential Solutions



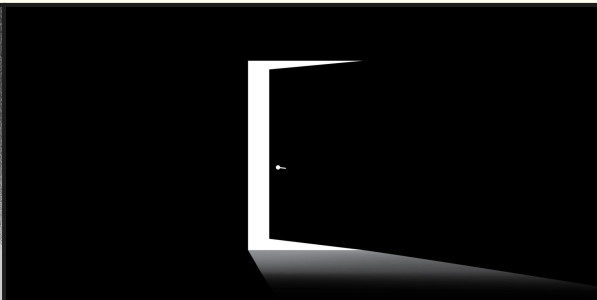
Speech Assessment Challenges

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Interrater variability

- Speech evaluation is inherently subjective
- Trained raters differ in applying scoring criteria
- Performance standards are complex and difficult to operationalize consistently



Limited independent validation

- Largely based on human-machine correlations
- Evidence often produced by test developers
- Rater characteristics and procedures are underreported

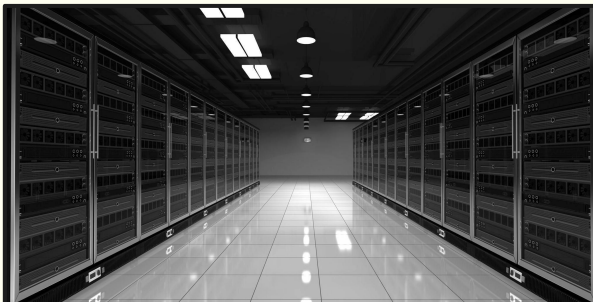


Accessibility constraints

- High cost and dependence on trained raters
- Limited interpretability of scores for teachers and learners

Speech Assessment Solutions

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Commercial automated systems

- Widely deployed (e.g., SpeechRater, Versant, DET)
- Evolved from feature-based models to neural systems
- Scores derived from statistical mappings to human ratings

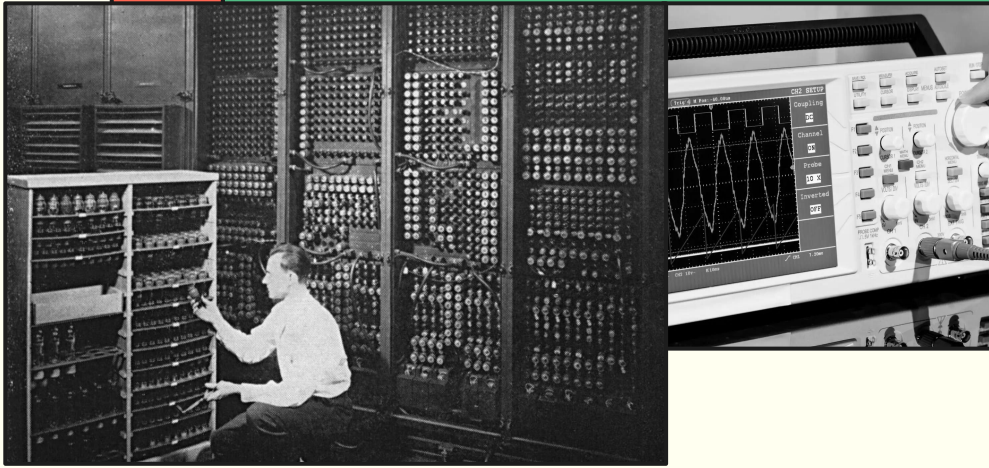
Experimental automated systems

- Machine learning models approximate human speaking judgments
- Use of hierarchical and transformer-based architectures
- Strong predictive performance within datasets

Azure pronunciation assessment

- Production-level system for Accuracy, Fluency, Prosody
- Combines speech recognition features with multi-level modeling of speech patterns
- Evaluated primarily through correlations with expert ratings

Methods & Measurement



The Datasets

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International Corpus Network of Asian Learners of English (ICNALE)

3,510 spoken monologues from learners across 10 countries, with CEFR levels (A2-B2+) assigned using external standardized benchmarks (e.g., TOEFL, TOEIC, IELTS).



Classroom contexts:

University EFL settings in China, Korea, and Japan using a controlled read-aloud task (Please Call Stella) under typical instructional conditions.

Classroom participants:

66 university-level EFL learners from China (28), Korea (20), and Japan (18).

Classroom raters:

Three experienced EFL professionals assigned scores on a calibrated 0-100 scale following a structured calibration procedure.

Linguistic Indices

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TAALED

Tool for the Automatic Analysis of Lexical Diversity: Provides indices of lexical diversity reflecting the range and distribution of vocabulary use (e.g., MTLT, HD-D).

TAALES

Tool for the Automatic Analysis of Lexical Sophistication: Captures multi-dimensional lexical sophistication, including frequency, range, and age of acquisition.

TAACO

Tool for the Automatic Analysis of Cohesion: Quantifies discourse cohesion by capturing how ideas are linked through lexical overlap and semantic similarity.

TAASSC

Tool for the Automatic Analysis of Syntactic Sophistication and Complexity: Provides fine-grained indices of syntactic sophistication and usage-based grammatical patterns.

L2SCA

L2 Syntactic Complexity Analyzer: Operationalizes syntactic complexity using clausal and phrasal structural measures.



Prosody, Accuracy & Fluency

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Prosody

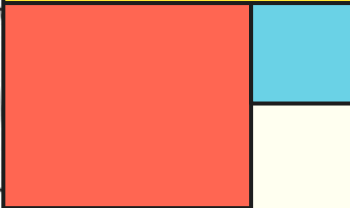
- Stress, rhythm, and intonation
- Helps listeners process meaning

Accuracy

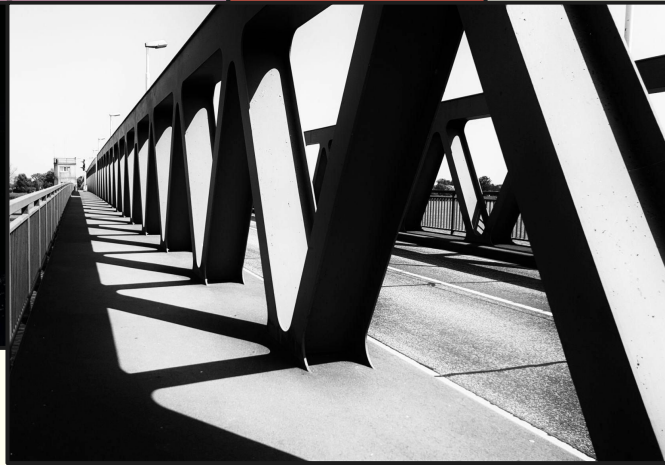
- Segmental control
(vowels, consonants, clusters)
- Contrasts necessary for intelligibility

Fluency

- Temporal control (rate, pauses, repairs)
- Reflects processing efficiency



Results & Future Directions



Correlates & Predictors

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CEFR Correlates

1. **TAALED**
Basic Ntypes
2. **AZURE**
Prosody Score
3. **AZURE**
Overall Pronunciation
4. **TAALED**
Basic Nfunction Types
5. **TAALED**
HDD42 AW
6. **TAALES**
Kuperman AOA TP AW

CEFR Predictors

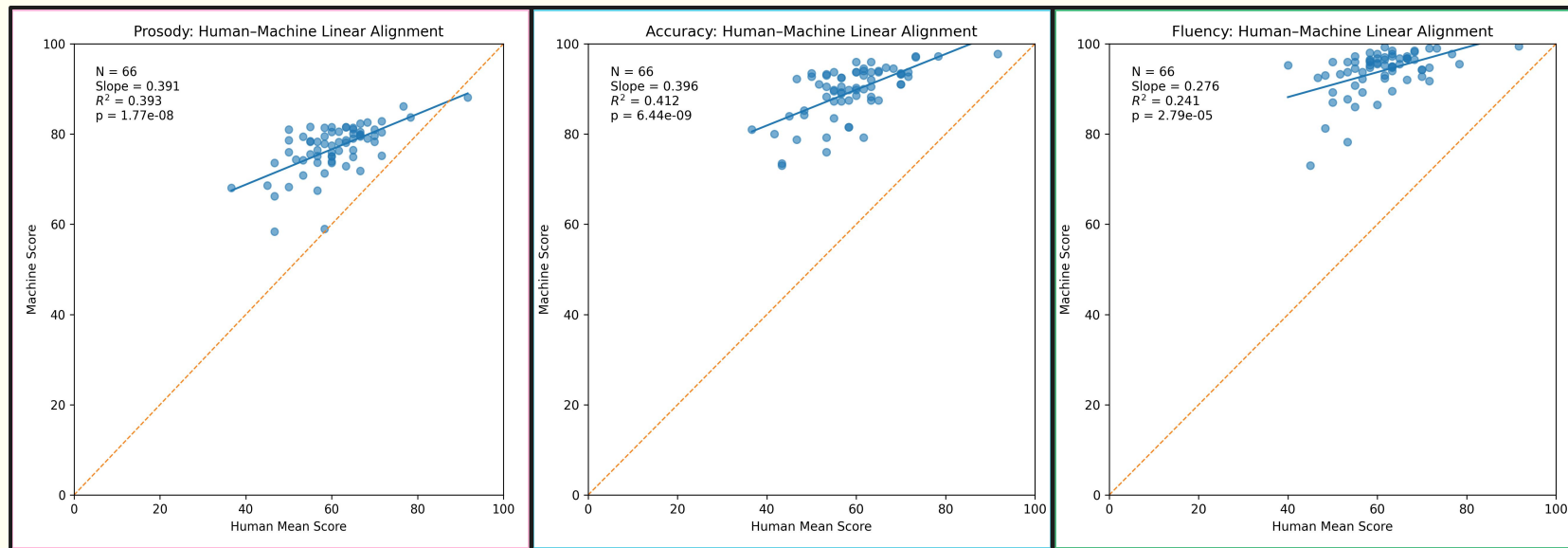
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Basic Ntypes
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Basic Nfunction Types
3. **AZURE**
Prosody Score
4. **TAACO**
Function MATTR
5. **TAALED**
HDD42 AW
6. **TAALED**
Basic Ntokens

Note:

These are $p < .001$ Spearman's Correlation & Ordinal Coefficient statistics relating measures of speech production and CEFR proficiency level for the ICNALE dataset. For spoken monologue data, the number of words speakers could spoken in a minute was the strongest CEFR correlate/predictor. Azure pronunciation measures were nearly as strong.

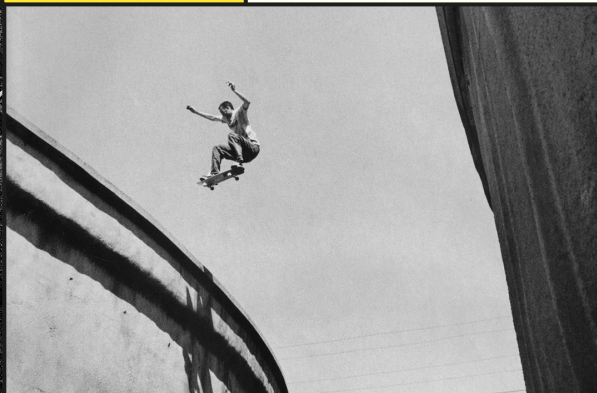
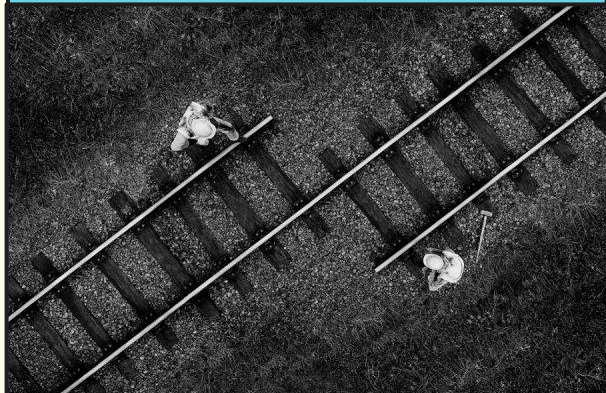
Construct Validation

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Discussion

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Signal, Scale, and Fluency

- **Strong rank-order agreement** (robust re: accuracy & prosody)
- **Systematic bias** (misalignment between human & machine scores)
- **Fluency measurement issues** (rater variability & misalignment)

Limitations & Future Research

- **Monologue task only** (extend to diverse/interactive tasks)
- **Human rating variability** (refine constructs & link scores)

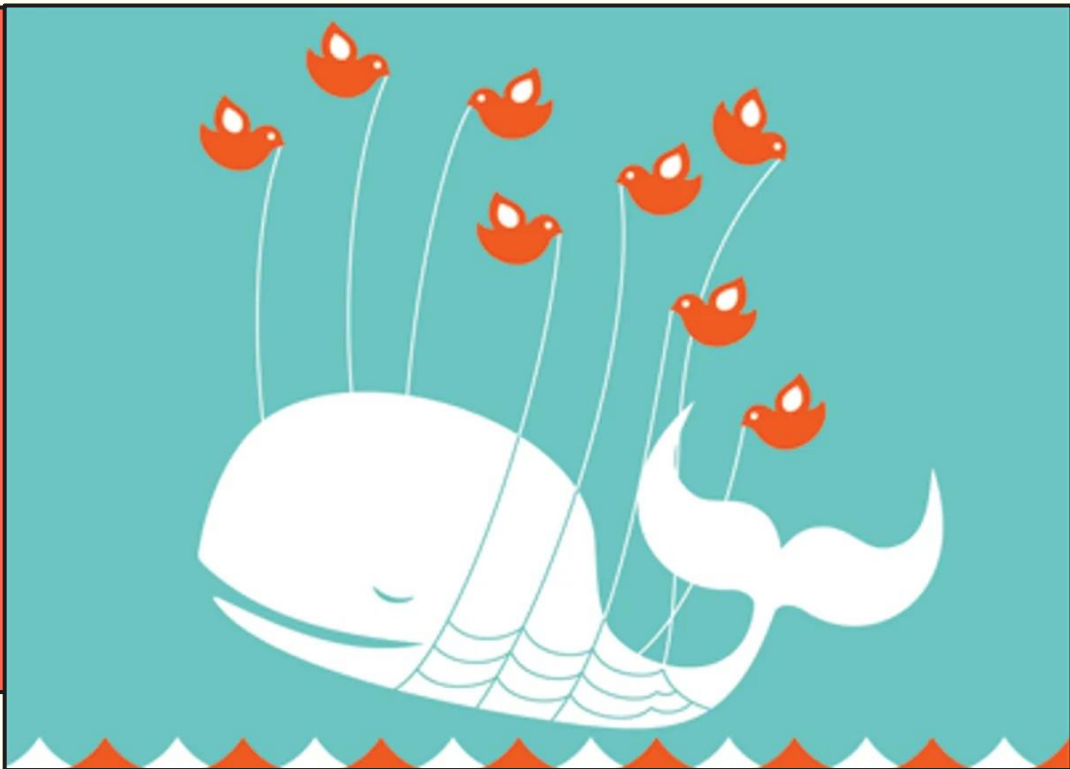
Implications for Practice

- **Best suited for formative use**
- **Supports ipsative tracking** (change over time)
- **Useful accuracy & prosody feedback**
- **Requires calibration or human oversight for high-stakes use**

Live Speech Test Demo

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- Let's check the results of the demo!
- [Demo Link Here!](#)
- [Demo Results Here!](#)
- [Backup Results](#)
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Q&A

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Citation:

Rose, R., Gaynor, B., & Fowler, J. (2026).
An Argument-Based Validation of
Azure Pronunciation Assessment
Across Corpus and Classroom Data.
Manuscript in press.

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